MSX and SPECTRAVIDEO

COMPUTER FORUM

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Magazine

SUPER SOFTWARE SPECIAL



MSX SOFTWARE AND HARDWARE

Unfortunately shortage of time has prevented me from preparing my advertisement this month, and publishing deadlines must be met.

I have several new pieces of MSX software in stock as well as some new hardware items. To obtain a price list send a Stamped and addressed 9" x 4" envelope with a note to:

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MSX and SPECTRAVIDEO
COMPUTER FORUM
Magazine

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EDITORIAL

Well, the Perth Electronics Show has been and gone. On display were the usual TV receivers, LOUD stereo equipment and the very latest in burnt offerings from the microwave oven people.

Where were the computers? There were a couple of stands showing IBM clones, the usual Commodore and Microbee crowd and a few others. MSX was there — exactly three machines. Toshiba had their HX10 and Sanyo were displaying the PHC27 and PHC33. Sony had everything but computers. AWA Thorn, who distribute the Mitsubishi MSX and Amstrad, didn't have a stand at all. As for the other MSX brands — JVC, National, Yamaha, Spectravideo, Canon, Pioneer, Philips, Sharp, Hitachi, Casio, Fuji and General — well, those that had stands must have forgotten that they manufacture MSX computers and the rest forgot to even book a stand.

It strikes me that the distributors of MSX computers in Australia have a great deal of faith in MSX. So much faith, it seems, that they think it will sell without even telling people about it.

Happy computing ...

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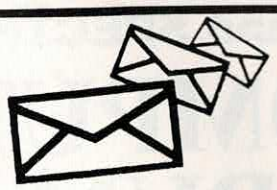
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MAILBOX



This page is for your comments and queries. If you have a specific problem or have solved one, we would like to hear about it.

If you require an answer before we publish it, please enclose a stamped self-addressed envelope.

UPDATING TO MSX

Please find attached a subscription for your magazine. Any magazine of this type is very much appreciated and I wish you good luck in your endeavours. I own a SV328 with 605 Expander, two double-sided disk drives, RS232 and Modem and 80 Column Card.

I, like many others, am interested in MSX but I am not willing to outlay many hundreds of dollars to change over when I have a very good system already operating. To assist with updating I am very interested in the MSX adaptor — where can it be obtained? How much? Is it any good?

I am not a very good programmer and some programs I would like to see published are —

1. Tape cassette program for SV318/328 to read MSX Tapes.
2. Hex. Find program to list port In and Out's of an MSX program so they can be manually converted to suit a 318/328.

Is there a BASIC that can be loaded into one of the other pages (of RAM) to make the 328 look as much as an MSX machine as possible? Also is there a tape BLOAD program to read the Hex addresses of a tape so that it can be put onto disk?

If you or any of your readers can assist with the above I would appreciate it very much. I am even willing to purchase some of the above programs if they exist.

Kevin Callaghan
East Preston, Victoria.

The MSX adaptor would seem to be a logical acquisition for someone in your position. However, the only one that we have contact with belongs to one of our readers in NSW who purchased it in Hong Kong. Some SV dealers were stocking it, but, due to the fact that it reduced the usable RAM to 16K, it did not sell very well. We believe that Bondwell, who manufacture the SV are taking another look at it. Why must everything be so complex?

Our resident hacker, Robert Brinkworth, is looking at some of the software you mention, and we hope to publish those types of utilities in future issues of Computer Forum.

CX5M SHORT ON MEMORY

I own a Yamaha CX5M Music Computer and was told by the dealer that all MSX machines were compatible. Several months later I bought "The Hobbit" and was unable to load it into the CX5M. I was told my present tape deck was not compatible. I bought a new tape deck and still could not load the tape. I was then told that the CX5M did not have enough memory — it is only a 48K machine and would have to buy a 64K memory expander.

Being somewhat sceptical of a salesman's advice, I am now attempting to do some homework. If I put in this program:

```
10 For I= -32768 TO 80000
20 PRINT I ; PEEK(I)
30 NEXT I
```

overflow does not occur until 65536. Doesn't this mean a memory size of 64K? What is the significance of the negative numbers? I thought that perhaps the music functions occupy too much memory and that there may be a way to clear this information temporarily to allow the loading of larger programs. I can make no sense out of the distributors and they cannot even provide me with a memory map.

Andrew Arthur
Templestowe, Vic.

Although there is information available about the Yamaha CX5M as a music computer, the actual MSX side has been largely ignored.

MSX computers are sold variously as 32K, 48K, 64K and even 80K (in this case they count the 16K VRAM). What your program does is read the upper 32K MEMORY ADDRESSES. This does not necessarily mean that the RAM is accessible or even fitted. Remember that you can even Peek at ROM locations, but Pokeing will have no effect. You could modify your program to Poke the address with a fixed value and then Peek to see if it is the same, in which case RAM is present. This can be dangerous if you poke around the area where your program is stored or the System area.

A far simpler way is to type NEW and press ENTER then type PRINT FRE(0) and press ENTER. The value displayed is the actual USABLE RAM. Programs like "The Hobbit" require approximately 26K of usable RAM. In your program, the negative number simply means 65536-32768. The loop is actually starting at 32768.

We suggest you contact Erik Kowarski, C/- MSX User Group, P.O. Box 87, Mt Hawthorn, W.A. 6016. He also has a CX5M and may be of further assistance.

CASSETTE LOADING PROBLEMS

Thanks for publishing a magazine about computers using every-day jargon that even I can understand. Recently I have "updated?" my Oric ATMOS and purchased a SONY HIT-BIT MSX.

The Sony is a superb machine to use but I just cannot get it to CLOAD at the fast baud rate of 2400. There is no problem CSAVE-ing to tape; giving a clear, crisp signal. The recorder is an AIWA and we have no problem using it on the ORIC at 2400 baud. I have tried the following commands.

SCREEN,,,2:CLOAD
and LOAD"CAS:[file name]"
and all variations of these. The computer goes into the loading state; but just doesn't load.

To get a program to automatically start after loading from tape, the command RUN"CAS:[file name]" works OK; but to load this from tape the program needs to be saved on the tape using the command SAVE "CAS:[file name]". Now, on my HIT BIT, when saving in this manner the computer keeps turning the recorder off and on every few seconds as if it is saving the program in batches. Is this normal? If so, it wouldn't be so bad at a higher baud rate; but it takes forever at 1200.

Always in the past when I have been convinced that the computer is faulty because it won't work as I think it should, by further study and perseverence on my part the computer has proven faultless. I hope it is the case this time and you may be able to put me straight.

Colin Johnstone
Chadstone, Victoria

Firstly, Colin, we hope that the cassette explanations in this month's tutorial will be of assistance. You are quite correct in your assumption that the SAVE command transfers the program in batches — 256 bytes at a time, to be exact. This is slow because the program is saved in ASCII format and not compressed. The computer saves a header signal at the start of each block as well. The baud rate for SAVE is fixed at 1200 for some reason — possibly for purposes of reliable transfer. Pity!

Turning to your first problem, the loading baud rate of programs is fixed at the time the particular program is saved. That is, a code corresponding to whichever 'tail' was on the SCREEN or CSAVE command at time of transferring the program is dumped to cassette along with the program. Therefore a 2400 baud program cannot be loaded at 1200 and vice versa.

We are assuming that when you enter CLOAD you are trying to load a program that was CSAVED. Also, if there is any doubt about the exact "file name" (even lowercase will make a difference), use LOAD"CAS:" for the LOAD command. It may seem a stupid comment, but we hope you are not entering the square brackets with the file name. We do not know which model AIWA recorder you have, but if there is a volume control fitted, try cloading at different settings.

DRAW BUGS IN SV?

My congratulations to you for a most needed magazine. I have an SV328 disk system in which there is a "bug" in BASIC, namely the rotation command in the graphic language. The 0 and 180 degree modes (A0 and A2) work correctly, the 90 and 270 degree modes (A1 and A3) result in a syntax error. I wonder if anyone else has this problem and, hopefully, knows a solution.

Also, I am in need of the circuit diagrams for the SV801 disk interface as mine has an intermittent fault. Circuits of the 64K expansion card and the 80 column card would complete hardware details of my system and may prove useful in the future.

As soon as time permits I intend completing circuits for a battery-backed real time clock and possibly a voice synthesiser to fit the SV601 expander. If you think circuits and driver software for these would be of interest to your readers I will send them to you when available.

Finally, I have a short program for SV328 users which allows BASIC to utilise that "wasted" 32K RAM which is mapped out in order to access the BASIC ROMs. A machine code routine allows a specified part of high memory (32K to 64K) to be copied to low memory (0 to 32K), and vice versa.

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Simply run the BASIC program to load the machine code into protected high memory. To use the routine from BASIC just poke a start address of the source memory into locations 52006/52007, and the destination start address into 52009/52010, also the quantity of data into 52012/52013. Note that the most significant byte is the higher location of each pair.

BASIC code needed to access the routine is —

```
100 DEFUSR (0) = 52001
110 OUT 136,15
120 X = USR0(0)
```

The main program is —

```
10 CLEAR 300,52000
20 FOR I = 52001 TO 52020
30 READ C: POKE I,C
40 NEXT I
50 END
60 DATA 62,221,211,140,33,0,0,17
70 DATA 0,0,1,0,0,237,176,62,223
90 DATA 211,140,201
```

Don Mayfield
West Pennant Hills, NSW

Thanks for the routine, Don. We have not had time to check it out, but I am sure our readers will do so.

We certainly would be interested in any circuitry and driver software which you can supply for the projects you mention. The problem you are having with the ANGLE command would seem to be peculiar to SV318/328. The following example demonstrates how to simulate the problem —

```
10 SCREEN 1: REM MSX USE SCREEN 2
20 DRAW "BM100,100"
30 FOR I = 0 TO 3
40 DRAW "A = I;R20"
50 FOR T = I TO 100:NEXT T
60 CLS:NEXT I
70 GOTO 30
```

This works, but change the R20 to L20 in line 40 and see what happens. MSX works, but SV318/328 will not.

DEAD KEYS

I own a SV328. Could you please tell me how I can get the keys SELECT and PRINT to work? I have tried everything.

I think that you should print people's addresses and names, so that readers can write to each other.

S. Dimitrio
Footscray, Vic.

SELECT and PRINT are, in fact, DEAD keys. They can only be used by Peeking their RAM locations. These are accessed on the SV318/328 using Peek(&hFD83) — if Peek(&HFD83) = 223 if the PRINT key is pressed. For MSX, Peek(&HFBE1) = 191 for the SELECT key.

We can only print the full names and addresses of those people who request it. Maybe we could have a Pen-Pal section in future issues — what do you think?

MSX CONVERT

I am one of the founding members of the (Latrobe) Valley CoCo Nut User Group (Tandy Colour Computer) which we started in '82 along the strict guidelines of the late Greg Wilson Rainbow Network. Our group was composed of owners of various 6800 & 6809 type computers.

Over the past two years I have personally been following the MSX scene very closely, reading as much as I could — seeing as it was, in my opinion, the best BASIC and Sprite home computer on the market, making the Z80 (industry Standard Chip) work to its utmost. I am one for standardisation — hardware, software, I/O and interfacing, but it must be upwardly compatible so that research and development continues and MSX fits this picture.

But my, in Australia — poor sales promotion and marketing?? Darn shame. This is where your magazine should fill a vacuum on the Aussie market, so I wish you and your staff all the very best of success and will try and provide information from time to time.

As many Spectravideo MSX, X'press and Sony MSX machines have been sold here in Eastern Victoria, I thought the best thing to do my bit is start an MSX Users Group and have bought myself an MSX computer with disk drive.

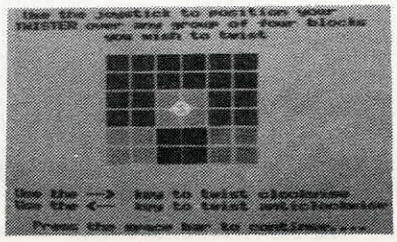
George Francis
Morwell, Victoria

Spot on, George. The MSX is the best thing since sliced bread and will far exceed sales and popularity of any other computer if it is promoted and supported in the same way that the Commodore has been. The MSX computers themselves are vastly superior to that machine which, in my opinion, was sold purely on the basis of massive sales promotion and a flood of software and information. We are trying to do our bit by showing confidence in MSX. Australian distributors please note — THE BUCK STOPS WITH YOU.

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Software NEWS & REVIEWS

FDS SOFTWARE (MSX & SV Tape)
SVPAK23/MSXPAK23
by Thinking Man's Software



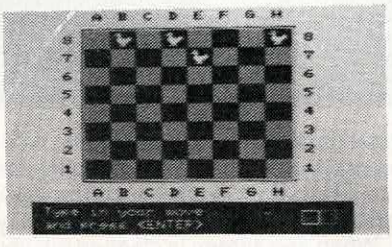
This is a two-game pack. The first title is TWISTER which presents you with 36 squares made up of groups of four squares, each group having a different colour. All 36 squares are then jumbled and your task is to re-arrange them into the original pattern. This is done by rotating blocks of four squares with your Twister. This requires planning and patience.

The other title is REVERSE. The numbers 0-9 are displayed on the screen in jumbled order. You must arrange the numbers in their proper order by reversing blocks of them.

Both come with complete instructions and have colourful displays.

Price: $16.00

SVPAK24/MSXPAK24
by Thinking Man's Software



Another two-program pack, this one contains FOX & GEESE and THINK. The first one is an adaptation of a board game where you must get the fox past the line of geese before he is cornered. Simple? Not quite. There are three levels of play — easy, difficult and almost impossible.

THINK is a type of word mastermind. You must guess the four-letter word that the computer has selected from the vast number stored in memory. All are valid four-letter words and all are clean!

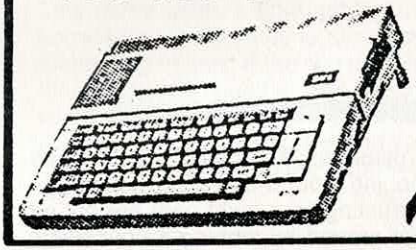
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UTIL 2 - two disk utilities: newdsk & llist; 2 very useful programs 1
MT-BASE - a user friendly ROM based database with many features 2
"recommended whole heartedly" by Canadian Users Group.
AEROBAT - a 3D graphics Cessna 152 flight simulator; does aerobatics!
1986 NZ Software Awards finalist!! MSXonly *3
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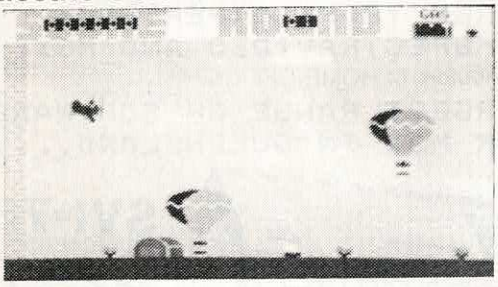
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MSX SUPER SOFTWARE SPECIAL

Our thanks to Peter Fox of Peter J. Fox (Computers) for supplying the software included in this review.

BARNSTORMER (MSX Tape)
by Electric Software

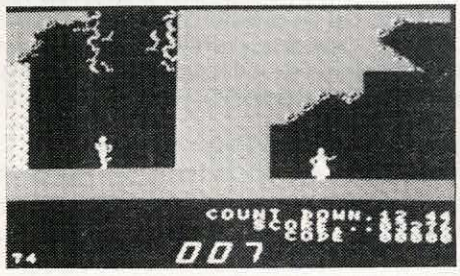


You are Barney the barnstormer. You must fly your plane across the screen dropping flour bombs on just about everything in sight. Obstructing your flight path are hot air balloons which you must ground by lobbing a flour bomb into the basket. Score points for bombing the barns etc along the way, but watch out for the clouds turning dark blue, because these give off lightning bolts which will knock out your plane.

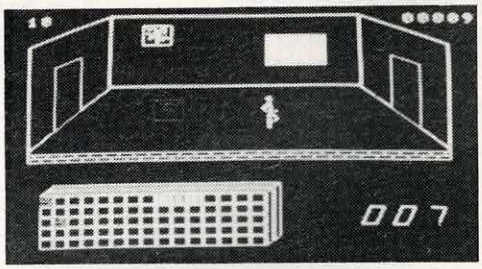
Once you have grounded all of the balloons, you will have a clear run up to the water tower in an attempt to penetrate through it. The plane gathers momentum moving across the screen, and this can make it difficult to control in tight spots. Good graphics and sound. Good value.

Price: $19.95

A VIEW TO A KILL (MSX Tape)
by Domark



Actually two games in one. The first of these is THE MINE — Silicon Valley. You are Bond — James Bond — and your mission is to defuse the nuclear detonator which has been placed in an abandoned mine under Silicon Valley. Items of use which can be found around the mine are a grapnel gun, dynamite, lighter and code numbers which you will require in the correct sequence to defuse the bomb.

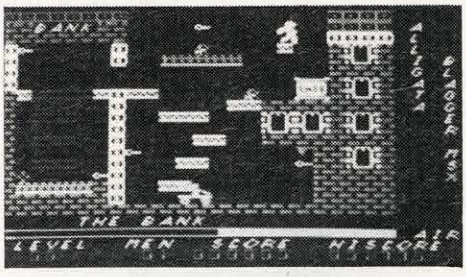


Game two is THE CITY HALL ESCAPE. In this adventure, Bond and his companion are trapped in the City Hall which has been set on fire by Max Zorin. Once again there are several useful items that can be found to aid in escaping the burning building. A side view of the building shows where the fire is and the rooms that Bond has been in.

Both games are a little difficult to get used to which will probably add to their lasting interest. Good graphics and the theme music is particularly good.

Price: $27.95

BLAGGER — MSX tape
by Alligata



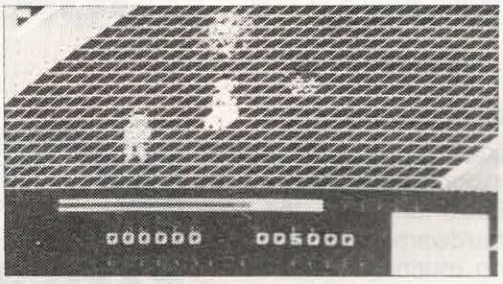
You are Roger the Dodger, master burgler, working your way through numerous buildings — searching for the golden keys, robbing safe after safe but always on the alert to the many obstacles and alarm systems, and those spooky, cooky night watchmen. When one screen is cleaned out you move to the next level in an endless pattern of banks, shops and houses.

There are 20 screens and on each screen you must collect all of the keys before you can open the safe. There are conveyor belts and ledges, some of which crumble away if you stay on them too long.

The graphics are quite well done, particularly Roger, and the movement is smooth. Theme music is from the Pink Panther and fortunately can be switched on or off from the keyboard without affecting the other sound effects.

Price: $19.00

DISC WARRIOR (MSX Tape)
by Alligata



No, this is on tape, not disk. Travel the 26 zones of the 3D computer complex alone on constant alert for the android guards and robot canines. Your mission is to destroy the master CPU. Armed only with your electrified "boomerang" power disc, blast your way through force barriers, defend yourself against enemies whilst searching out parts of the master key and constructing the super bomb before the ultimate destruction can be achieved. But keep a watch on time — your energy level is constantly increasing, but each encounter saps your strength, overactivity will leave you stranded in this alien world.

Game plan: Travel the zones on foot and by travel disc (boarded from walls), collect the coloured door keys, parts of the master key and superbomb. When all are obtained, final destruction can be made. Energy cells will sustain energy level.

The 3D effect is quite difficult to get used to at first. Good graphics and sound effects. Would take a while to master. Requires good reflexes and joystick skills.

Price: $19.95

737 FLIGHT SIMULATOR (MSX Tape)
by Mirrorsoft



This is a flight simulator of a 737 type jet airliner with full instrumentation and out-the-windscreen view.

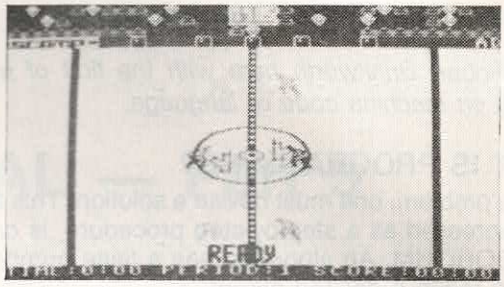
If you are looking for a dog-fight simulator, this will be disappointing because it has similar responses to a real passenger aircraft — fairly sluggish. In fact, you tend to "feel" the weight of the aircraft as it slowly responds to the controls.

Instrumentation is complete and quite easy to read as most are digital display. The forward view graphics are limited to only the runway. This view changes after takeoff to a plan view of the area showing runway and marker beacons. This program comes with a 28 page manual containing instructions and reasons why the aircraft behaves in a certain way.

This one is not for the arcade game or shoot-em-up fiends — more for the person wanting a simulator which is close to the real thing which requires a lot of calculation and judgement.

Price: $37.95

SLAPSHOT (MSX Tape)
by Anirog



A one or two-player game of ice hockey. The rules are the same and there are three three-minute periods in the game. You begin with three players per side — one team is red and the other green.

The active player of each team is the darker shade of his team's colour. He must take possession of the puck, skate down the ice to the opposing goal and shoot the puck into the net. The opposing team will attempt to steal the puck from your player by using skill or brute force. If the brute force, or "checking" is too severe, then a penalty will be alloted.

This game takes a little getting used to, particularly as a knowledge of the rules is assumed. However, this can be interesting in itself if the two players have the same level of ignorance of the game. The playing field, or rink, is three screens wide and scrolls left and right to follow the play. The 3D graphics are quite good and each goal is followed by a slow-motion replay and the faces in the crowd change as they cheer.

Price: $19.95.

STAR BLAZER — (MSX Cartridge)
by Sony



My first impression of this game was — how simple!

Guide your battle plane up and down as it moves across the screen and drop bombs on the enemy radar station.

Firstly, you must be below a certain altitude before you can drop your bombs. Pressing the trigger above this point only fires your missiles. You must maintain altitude as the plane has a natural tendency to sink toward the ground. Watch out for the towers which crop up at various intervals — you must climb to get above them.

The radar only appears every fourth or fifth screen and your fuel is getting low. You can be replenished with fuel and bombs by parachute from your supply plane which flies overhead every now and then. Watch out for the big bird which swoops in and carries off your parachute and supplies.

There are five missions, the first of which is bombing the radar. The graphics on this cartridge are simple but well defined. The use of sound effects is good, but a great deal more could have been done in this area. The saving grace is that it is a cartridge — just plug it in, switch on and go.

Price: $29.95

LEARN MACHINE CODE

WITH ROBERT BRINKWORTH

Hello, Robert Brinkworth here with the first of many tutorials on machine code or language.

WHAT IS PROGRAMMING?

Given a problem, one must devise a solution. This solution, expressed as a step-by-step procedure, is called an ALGORITHM. An algorithm has a finite number of steps to reach a desired output.

An example of an algorithm is that of opening a door —

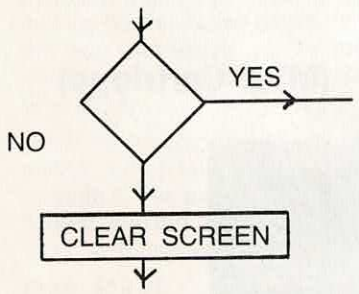
1 — insert key in keyhole
2 — turn key one full turn to the left
3 — seize doorknob
4 — turn doorknob left and push the door

The desired output being that of the now opened door.

This is a rather obscure analogy to that of programming a computer but indicates that step-by-step commands or instructions must be made in order for the program or sequence to operate as planned. Converting an algorithm to a sequence of instructions is called PROGRAMMING.

FLOWCHARTING

Flowcharting involves writing out systematically a series of diagrams with symbols. For example —



A test is done and is branched off accordingly.

A command is executed.

On this subject I have mixed emotions since I feel that flowcharting using diamonds and rectangles is a waste of time, but then I don't discourage writing down the proposed sequence.

Explaining what I have just said —

Object: To keep a room at a certain temperature.

step	operation
1	read temperature
2	compare with desired temperature
3	go to 6 if temperature too high
4	go to 8 if temperature too low
5	temperature OK go to 1
6	turn off heater
7	go to 1
8	turn on heater
9	go to 1

If you write out your program in this way then you can instantly see what routines must be provided for the program.

Until next time ...

BOOKS • BOOKS

MSX EXPOSED

by Joe Pritchard



This book is a very useful addition to your bookshelf. It covers the MSX Basic language, the standard hardware and some interesting routines for accessing the VDP and PPI.

The first three chapters cover the Basic commands in alphabetical order, basic variables, array variables, expressions and defining functions.

The next five chapters explain cassette loading, saving and handling files. The ON commands, VDP screen modes, graphics commands, sprites, joysticks and the sound commands are also covered.

The final four chapters get a little heavier and discuss controlling the PPI, MSX memory map and storage of variables, sample routines and some information about MSX machine code. An appendix explains the number systems used by MSX.

Review copy supplied by Peter J. Fox Computers. Retail price $19.95.

Publisher: Melbourne House
227 pages

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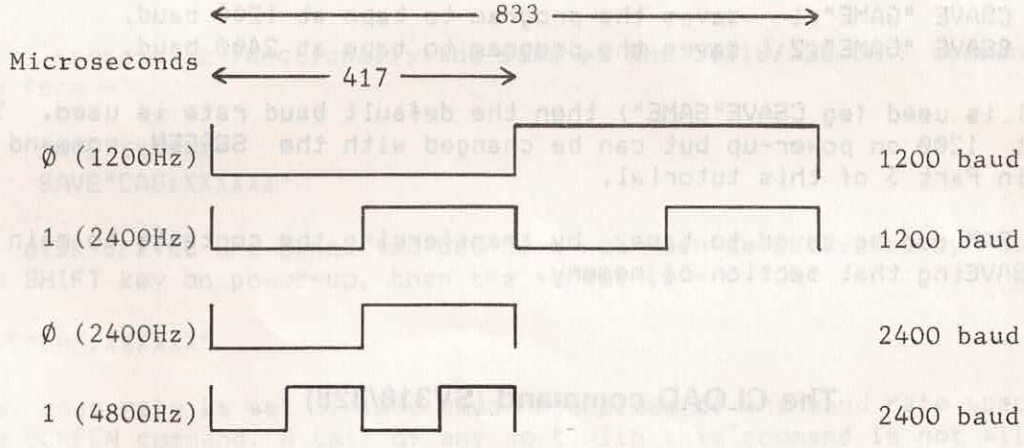
BASIC TUTORIAL — Part 5

LOADING AND SAVING PROGRAMS ON TAPE

This month the Basic Tutorial will deal with the commands that control the inout and outout of your computer. These commands cover the areas of saving and loading programs and storing and retrieving information on cassette. At a later stage, we will explain the same commands for use with disk systems.

In this section you will be confronted with the term BAUD or BAUD RATE. These are the terms used to describe the speed at which the information is transferred from the computer to the cassette or cassette to computer.

The BAUD or BAUD RATE is expressed in Hertz (Hz) or Cycles Per Second (CPS). They both mean the same thing - ie 1HZ = 1CPS. For the technically minded, this means that at 1200 baud, a 0 is a 1200hz signal and a 1 is a 2400hz signal. At 2400 baud, the 0 is 2400hz and a 1 is 4800hz.



The program is recorded onto the tape as a leader, followed by the actual program. This leader contains the name of the program and the baud rate. If you play a program tape on a tape system with volume meters, you will see the meter show a reading as the leader is read, then drop to zero before reading the rest of the program.

NOTE: With CSAVE, SAVE and BSAVE, the tape recorder must be in record mode BEFORE pressing the ENTER key. The computer starts sending information to the cassette as soon as this key is pressed.

The CSAVE command (SV318/328)

This command is used to save the BASIC program that is currently in memory to the cassette recorder. It takes the form -

```
CSAVE "XXXXXX"
```

The six Xs in quotes can be replaced with your own name for the program but you must not exceed six characters. If you do , it will ignore the extra. For example if you use CSAVE "NAPOLEON" the program will be saved as NAPOLE - however if you enter CLOAD"NAPOLEON" the computer will find NAPOLE on the tape and load it. In other words, the computer only looks for a match on the first six letters.

The CSAVE command can also be in the form

```
CSAVE "XXXXX",S
```

If the comma and S are used, the computer will save to cassette the entire contents of the screen memory or VRAM.

The baud rate cannot be altered for the Spectravideo and is set at 1800. This is why an MSX tape cannot be read by an SV computer.

The CSAVE command (MSX)

This command is the same as the one for SV318/328 except that the ,s option cannot be used. In its place is the baud rate option.

```
Example:  CSAVE "GAME",1   saves the program to tape at 1200 baud.
          CSAVE "GAME",2   saves the program to tape at 2400 baud.
```

If no tail is used (eg CSAVE"GAME") then the default baud rate is used. This is set at 1200 on power-up but can be changed with the SCREEN command as described in Part 3 of this tutorial.

The Video RAM can be saved to tape, by transferring the contents to main RAM and then BSAVEing that section of memory.

The CLOAD command (SV318/328)

This command is used to load a BASIC program from cassette which was saved using the CSAVE command. It takes the form -

```
CLOAD "XXXXXX"
```

The six Xs can be replaced with the name of the program - if known. For example - CLOAD "GAME". The computer will read the cassette looking for a program with the name GAME which, when found will be loaded. The name used must be EXACTLY the same as the one used when the program was CSAVED. In this example, if you used "game" and the program was saved as GAME then it will not be loaded.

If doubt exists as to the exact name of the program, just use CLOAD and press enter to load the next program encountered on the tape.

The CLOAD command (MSX)

This command is the same as for the SV318/328.

You do not need to specify the baud rate when CLOADING, as this is sensed at the time that the computer reads the first part of the recording on the tape.

The SAVE command (SV318/328)

This command is also used to save the current BASIC program in memory to tape. It takes the form -

```
     SAVE "XXXXXX"
or   SAVE"CAS:XXXXXX"
```

The program name is limited to six characters the same as CSAVE.

This command saves the program in ASCII format and is done in segments of 256 bytes. That is why the recorder switches on and off during the save function. This command must be used to save a program which is to be used with the MERGE command which is explained later.

The SAVE command (MSX)

This command is functionally the same as the SV318/328 SAVE command and takes the form -

```
     SAVE"XXXXXX"
or   SAVE"CAS:XXXXXX"
```

If disk drives are connected but have not been de-activated by holding down the SHIFT key on power-up, then the format is -

```
SAVE"CAS:XXXXXX"
```

The baud rate is set at 1200 baud regardless of any baud rate specified with the SCREEN command. A tail of any sort with this command is not allowed.

The LOAD command (SV318/328)

This command is used to load a BASIC program which was recorded on tape with the SAVE command. It takes the form -

```
     LOAD"XXXXXX"
or   LOAD"CAS:XXXXXX"
```

The six Xs can be replaced with the program name, if known. If the program

name is not known then LOAD"CAS:" will load the next program found on the tape.

The LOAD command (MSX)

This command is functionally the same as the SV318/328 LOAD command and takes the form -

```
    LOAD"CAS:XXXXXX"
or  LOAD"CAS:XXXXXX",R
```

where CAS: indicates that it is to be loaded from cassette and XXXXXX is the name of the file (if known) to load. The ,R option will load and RUN the program which was SAVEd.

The BSAVE command (SV318/328)

This is the Binary SAVE command and is used to save a block of memory to tape in binary format. It takes the form -

```
    BSAVE"XXXXXX",number1,number2,number3
```

where the six Xs can be replaced with the name of your program (up to six characters in length).
number1 is the start address of the section of memory you want to save.
number2 is the end address of the section of memory you want to save.
number3 is the entry address if the block of memory contains a machine code program and the entry address is different from the start address.

number1 and number2 must be specified and be greater than 32767. The number3 is optional.

The BSAVE command (MSX)

This command is functionally the same as the SV318/328 BSAVE command. It takes the form -

```
    BSAVE"XXXXXX",number1,number2,number3
or  BSAVE"CAS:XXXXXX",number1,number2,number3
```

The BLOAD command (SV318/328)

This is the Binary LOAD command and is used to load a binary coded tape which can be a machine-code program or just a block of memory which was BSAVEd to tape. It takes the form -

```
    BLOAD"XXXXXX"
or  BLOAD"XXXXXX",R
```

The ,R option uses the start address that was saved as per number3 in the BSAVE command. If no value was saved then this option assumes that the entry

point to the machine code program is at the start of the block of memory as specified by the number1 in the BSAVE command.

A value can be given to offset the program to load to a different part of memory from where it was saved. For example

BLOAD"XXXXXX",R,200	loads the information from tape to a memory position 200 higher than when it was saved.
BLOAD"XXXXXX",R,-200	loads the information from tape to a memory position 200 lower than when it was saved.

The BLOAD command does not check for a usable area of memory and care must be taken that this function does not overwrite the System Area of RAM.

The BLOAD command (MSX)

This performs the same function as the SV318/328 BLOAD command. If disk drives are attached and not de-activated the format is -

BLOAD"CAS:"
or BLOAD"CAS:XXXXXX" where XXXXXX is the filename

TECHNICAL DIFFERENCES BETWEEN THE VARIOUS SAVE AND LOAD COMMANDS

CSAVE saves the BASIC program in tokenised format similar to the way it is stored in the computer's memory. In other words, the BASIC command DATA will not be stored on tape as D A T A but as the number 132 which is the token used for this command. The program is recorded on the tape as a header of a fixed length (this sounds like a high-pitched whistle when played on your stereo) with the program name attached and then the program itself is recorded in a continuous stream of ones and zeroes.

SAVE records the BASIC program on tape in ASCII format, using the ASCII code for each number or letter in a program line. This makes the recording appear to the computer as very similar to a data file. The program is recorded on the tape as a header of a fixed length with the program name attached. This is then followed by blocks of 256 pieces of data.

A program which was recorded with the SAVE command cannot be CLOADed and one recorded with the CSAVE command cannot be LOADed.

The MERGE command (SV318/328 & MSX)

Another command for getting a program from tape into memory is the MERGE command. Remember, we can only MERGE a program which was recorded on tape using the SAVE command.

The MERGE command format is the same as LOAD and can be used with or without the name of the program to be merged. If the name is not used, then the next program encountered on the tape which was SAVEd will be merged.

The uses for this command include merging a routine or section of code which you want to use in more than one program - eg Sprite data, or introductions.

The program already in memory must not contain any line numbers that are the same as the program which is being MERGEd, otherwise these will be overwritten with the lines from the program on tape.

The CLOAD? command (SV318/328 & MSX)

This command is called CLOAD VERIFY and can be used to check a program that has just been CSAVED. To use this command, CSAVE the program in the normal way and, while the program is still in memory, rewind the tape and type CLOAD? and press the play button on the recorder. Now press ENTER. The program will now be read from tape and compared with the one in memory. If there is a mismatch, then an error message is displayed.

SUMMARY OF SAVE AND LOAD COMMANDS

```
CSAVE"GAMES1"          - saves the program GAMES1 to cassette. In MSX, a baud
                         rate of 1200 is used.
CSAVE"GAMES1",S        - saves Video RAM to cassette as GAMES1. (SV only)
CSAVE"GAMES1",1        - saves the program GAMES1 to cassette at 1200 baud. (MSX)
CSAVE"GAMES1",2        - saves the program GAMES1 to cassette at 2400 baud. (MSX)
SAVE"GAMES1"           - saves the program GAMES1 to cassette in ASCII format.
BSAVE"GAMES1",35000,35200          - saves a block of memory from 35000 to 35200
BSAVE"GAMES1",35000,35200,35100    - saves a machine code program from memory
                                     locations 35000 to 35200, entry point 35100

CLOAD                  - loads the next program on tape that was CSAVEd.
CLOAD"GAMES1"          - loads the program GAMES1 from tape.
LOAD"CAS:"             - loads the next program on tape that was SAVEd.
LOAD"GAMES1"           - loads the program GAMES1 from tape.
LOAD"CAS:",R           - loads and runs the next program on tape that was SAVEd.
LOAD"GAMES1",R         - loads and runs the program GAMES1
BLOAD"GAMES1"          - loads a block of memory called GAMES1 to the same area
                         from which it was BSAVEd.
BLOAD"GAMES1",R        - loads and runs machine code program GAMES1
BLOAD"GAMES1",R,200    - loads machine code program GAMES1 to a memory location
                         200 higher than when it was BSAVEd, and runs it.
BLOAD"GAMES1",R,-200   - loads machine code program GAMES1 to a memory loaction
                         200 lower than when it was BSAVEd, and runs it.
MERGE"GAMES1"          - MERGEs the program GAMES1 with any program currently
                         in memory.
CLOAD?                 - compares the last program CSAVEd with the one currently
                         in memory.
```

LEARNING ABOUT NOTHING — The importance of zero

Zero (0) plays an important part in programming your computer. Just about everything starts with zero. The first memory location is not 1 but 0. The

first bit of a binary number is called 0. The first SCREEN mode is 0.The leftmost position on the screen in all modes is 0 as is the topmost position. Let us now look at some areas where this fact is not so obvious.

When using ARRAYS, we might dimension an array as DIM A$(10). At first, it would appear that the A$ array has ten elements (1 - 10). In fact it has eleven (0 - 10). A$(0) is a valid location.This can be very important when you are dimensioning several arrays and memory is getting short. Remember that when you DIMension an array, a section of memory is used up to allow for each element of the array.

When the computer first boots up, it assumes that you might want to use string variables and allocates 200 bytes for this purpose. Sometimes this can be insufficient and an 'Out of string space' error will result unless you CLEAR a greater amount of memory at the start of the program. If you are only working with numeric variables, then whatever string space has been CLEARed can be recovered by using CLEAR0.

GRAPHICS COMMANDS

This month we continue the Graphics Commands section of the tutorial with the LINE and DRAW commands.

The LINE command

This command is used for drawing lines between co-ordinates on the graphics screen.

Spectravideo SCREEN 1 and MSX SCREEN 2 are the high resolution modes and the co-ordinates are 0-255 across the screen and 0-191 down. Any values given outside this range are enterpreted as the nearest valid point on the screen. For example 326 will convert to 255 in the horizontal range and 191 in the vertical range.

LINE(50,100)-(360,100) will have the same effect as LINE(50,100)-(255,100) and LINE(50,50)-(50,360) will have the same effect as LINE(50,50)-(50,191).

Spectravideo SCREEN 2 and MSX SCREEN 3 are the low resolution screens and the screen is divided into 64 locations horizontally and 48 blocks vertically. Although the co-ordinates have the same range, each co-ordinate within that range is taken to the closest block of 4. That is, in the horizontal direction 0-3 are the same, 4-7 are the same and so on.

Try this example -

MSX version

```
10 SCREEN 3
20 OPEN "GRP:" AS 1
30 FOR I=100 TO 255
40 PRESET(20)-(I,20),1
50 PRESET(20,100):PRINT#1,I
60 FOR T=1 TO 100:NEXT T:CLS
70 NEXT I
```

SV version

```
10 SCREEN 2
20 FOR I=100 TO 255
30 LINE(0,20)-(I,20),1
40 LOCATE 20,100:PRINT I
50 FOR T=1 TO 100:NEXT T:CLS
60 NEXT I
```

The line command can take many forms -

`LINE (10,10)-(100,10)`	this is the minimum configuration and will draw a line in the current foreground colour.
`LINE (10,10)-(100,10),1`	this will draw the same line, but will use colour number 1 which is Black.
`LINE (10,10)-(100,100),1,B`	this will draw a box outline in Black with the top left corner at 10,10 and the bottom right corner at 100,100.
`LINE (10,10)-(100,100),1,BF`	this will draw the same box but will colour fill it in Black.
`LINE -(100,100)`	this will draw a line to location 100,100 from the last point specified by any of the graphics commands. If none has previously been specified then 0,0 is assumed.

The following program demonstrates variations of the LINE command in high resolution mode.

```
10 SCREEN 1,1  : REM for MSX use SCREEN 2,1
15 COLOR 15,4,4
20 LINE(10,10)-(100,10)
30 LINE(10,20)-(100,20),1
40 LINE(10,30)-(100,40),1
50 LINE(10,50)-(100,60),1,B
60 LINE(10,70)-(100,80),1,BF
70 LINE(10,90)-(100,90),1
90 LINE-(100,100),1
90 LINE-(10,100),1
100 GOTO 100
```

The DRAW command

The DRAW command is part of the GRAPHICS MACRO LANGUAGE and can also be used in high and low resolution modes.

The first instruction we will look at is the BM or BLANK MOVE instruction. This is used to give a starting point from which to draw left, right, up, down and diagonally.

This command has the same co-ordinate specifications as the LINE command and the same limitations in the low resolution mode. If the BM command is not used then, like the LINE command, the last point used by any of the graphics commands is used.

Other instructions are for the direction in which drawing will proceed. They are -

D = draw down	E = draw diagonal up and right
U = draw up	F = draw diagonal down and right
L = draw left	G = draw diagonal down and left
R = draw right	H = draw diagonal up and left

Each of these is used to prefix a number which indicates the number of dots or pixels (picture elements) to draw in that direction. For example -

D30 draws 30 dots down, R25 draws 25 dots to the right.

Each of these instructions will draw from the point specified by the BM instruction or from the finishing point of the last direction instruction - whichever was performed last.

This program demonstrates the DRAW command.

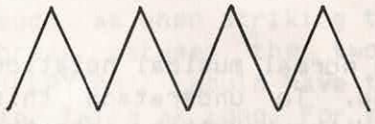
```
10 SCREEN 1,1 : REM for MSX use SCREEN 2,1
20 REM THIS DRAWS A CROSS
30 DRAW "BM50,50U20"
40 DRAW "BM50,50D20"
50 DRAW "BM50,50R20"
60 DRAW "BM50,50L20"
70 REM SO DOES THIS
80 DRAW "BM120,50U20D40U20R20L40"
90 REM AND SO DOES THIS
100 DRAW "BM50,120NU20ND20NR20NL20"
110 GOTO 110
```

SOUND COMMANDS

This month we continue the explanation and uses of the SOUND and PLAY commands.

Using the SOUND command

Registers 11 & 12 control the envelope period -



: EP : EP : EP : EP : - EP = Envelope Period

This can be demonstrated using the following program

```
10 SCREEN 2 : REM SV USE SCREEN 1
20 SOUND 7,62 : ' 00111110 sound on channel A
30 SOUND 0,0  : ' fine tune frequency of channel A
40 SOUND 1,8  : ' coarse tune frequency of channel A
50 SOUND 8,16 : ' turn on envelope
60 SOUND 11,0 : ' fine tune envelope period (can be 0-255)
70 SOUND 13,14: ' envelope shape
80 FOR I=0 TO 10 : CLS
90 A=I*3+4
```

```
100 LINE(E,120)-(A+A,90)
110 LINE-(A+A*2,120)
120 LINE-(A+A*3,90)
130 LINE-(A+A*4,120)
140 LINE-(A+A*5,90)
150 LINE-(A+A*6,120)
160 FOR T=1 TO 800 : NEXT T
170 SOUND 12,I : ' coarse tune envelope period
180 NEXT I : SOUND 12,0 : ' switch off
```

Using the PLAY command

NOTE: Before continuing with this section we must point out that there was an error in the explanation of the N command in the PLAY section last month.

The section should have read -

N number This will play the note given by the number from the table below. For example N55 will play the G note from octave 5.

01												02												03												04											
C	C#	D	D#	E	F	F#	G	G#	A	A#	B	C	C#	D	D#	E	F	F#	G	G#	A	A#	B	C	C#	D	D#	E	F	F#	G	G#	A	A#	B	C	C#	D	D#	E	F	F#	G	G#	A	A#	B
												1	1	1	1		1	1	1	2	2	2	2		2	2	2	3	3	3	3		3	3	3	4	4	4	4		4	4	4	5	5	5	5
0	1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7

05												06												07												08											
C	C#	D	D#	E	F	F#	G	G#	A	A#	B	C	C#	D	D#	E	F	F#	G	G#	A	A#	B	C	C#	D	D#	E	F	F#	G	G#	A	A#	B	C	C#	D	D#	E	F	F#	G	G#	A	A#	B
4	4	5	5	5	5	5	6	6	6	6	6	6	7	7	7	7	7	7	7	8	8	8	8	8	8	8	8	8	9	9	9	9	9	9	9	9	0	0	0	0	0	0	0	0	1	1	1
8	9	0	1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	9	0	1	2	3	4	5

Note: N36 is middle C on a piano.
N0 produces no sound.

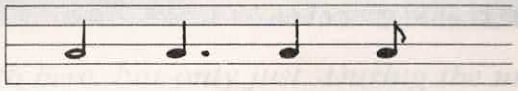
When using the N command, N0 will produce no sound. The lowest note that can be played with this command is C# in octave 1. However N96 can play one note higher than the highest note available with other methods - in effect C in octave 9.

Let us now look at how a piece of music is written in normal musical notation and compare that with how it looks as a Play command. To understand this section, you will need to know how music is written. Without getting into the realm of the music teacher, simply explained - music is written using five parallel horizontal lines called the Staff. The notes take their name from the point at which they appear on the staff in a vertical position.

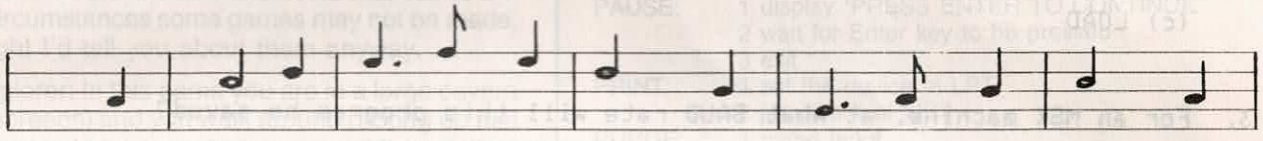
Starting with the bottom line, each line has the note E, G, B, D, or F. Notes falling between the lines fill in the gaps F, A, C, E.

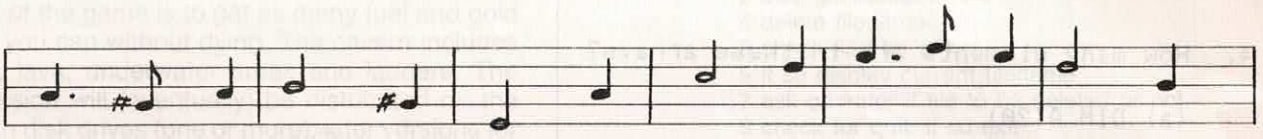


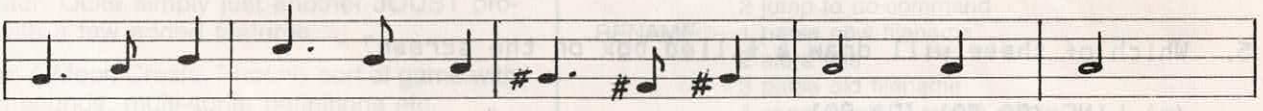
The length of a note is indicated by type of symbol used in that position on the staff.



The tune is Greensleeves -







First the musical notes are converted to their alphabetical equivalent. Then the Tempo command is added and the program looks like this.

```
10 PLAY "t160o4A4o5C2D4E3F6E4D2o4B4G3A6B4o5C2o4A4r64A3g#6A4B2G#4E2"
20 PLAY "o4A4o5C2D4E3F6E4D2o4B4G3A6B4o5C3o4B6A4G#3F#6G#4A2r64A4r64A2"
```

Where the same note is played twice in succession (as indicated by the *) - such as when striking the same key twice on the piano - then we must have a break between the two notes sounding. To simply use the A note twice (PLAY"AA") would have the same effect as holding down the A key on an organ for twice as long. For example -

PLAY"A4A4" sound exactly the same as PLAY"A2"

To overcome this we must use a Rest between the two notes - PLAY"A4r64A4"

Try to obtain some sheet music of a simple tune that you know well and have a go at converting it to work with the PLAY command.

QUESTIONS

1. What term is used to described data transfer rates?

2. To load a program which was CSAVEd but you have forgotten the name, which of these could you use?

   (a) CLOAD "??????"
   (b) CLOAD
   (c) LOAD

3. For an MSX machine, at what BAUD rate will this program be saved?

   SAVE"CAS:PROG"

4. How many elements are in these arrays?

   (a) DIM A(20)
   (b) DIM A$(10,10)
   (c) DIM A(5,5,5)

5. Which of these will draw a filled box on the screen?

   (a) LINE (50,50)-(80,80)
   (b) LINE (50,50)-(80,80),15,B
   (c) LINE (50,50)-(80,80),15,BF

6. LINE (0,0)-(255,191) will draw what?

   (a) A horizontal line
   (b) A vertical line
   (c) A diagonal line

7. What will be drawn with this command?

   DRAW "BM20,20R20D20L20U20"

8. Which octave contains N84?

9. Which of these notes plays the longest?

   (a) A3
   (b) B6
   (c) C4

10. How do you simulate the same note being struck twice?

HACKERS CORNER

Robert Brinkworth here, but only just. During the month my disk controller blew up! Word of warning folks — don't make connections to your disk drives with the power on ...

I must first say that I make no promises about the following games which I will describe. These are arcade games which I would like to make. Note that due to unforeseen circumstances some games may not be made, but I thought I'd tell you about them anyway.

Cavern Explorer: In this game you are in a large cavern (over 100 screens) and you walk around picking up fuel canisters and gold tokens which you must then deliver to a basement at the bottom of the cavern. Also, you have many monsters after you which must be destroyed. The object of the game is to get as many fuel and gold objects as you can without dying. The cavern includes conveyors, lava, underwater areas and ladders. The original version will (eventually) be distributed on the SV-328 with disk drives (one or more), later versions for the SV-318 and MSX computers will be made, that is after I finish writing the program.

Lance Master: Quite simply just another JOUST program but with a few added features.

Star Blaster: A Moon Cresta, Phoenix sort of game with nice added sounds, multi-sprite definitions etc.

THE REST OF THE DOS

AUTO:
1 parse for filename or command
2 put information in the BOOT files
3 exit

DIR:
1 parse for parameters
2 read DIRectory file
3 display file as per parameters
4 exit

COPY:
1 get all parameters
2 perform copy operation
3 exit

FORMAT:
1 get all parameters
2 perform formatting operation
3 exit

APPEND:
1 get both filenames
2 open first file
3 get EOF and NEXT pointers
4 open second file
5 copy contents of second file onto the end of the first file
6 close files
7 Exit

ATTRIB:
1 get filename and parameters
2 set file according to parameters
3 exit

FREE:
1 get parameters
2 select drive unit
3 count up free space
4 display it
5 exit

PDRIVE:
1 get parameters
2 set current drive/tape unit
3 exit

PROT:
1 get parameters
2 set disk parameters
3 exit

PAUSE:
1 display 'PRESS ENTER TO CONTINUE'
2 wait for Enter key to be pressed
3 exit

PRINT:
1 set the device to LPT:
2 jump to 2 of LIST

PURGE:
1 parse input
2 if null then display 'ALL (Y/N)?' message set flag to all if Y is pressed
3 else get filename
4 delete filename
5 check flag for all
6 if so display current filename
7 ask operator if file to be deleted or not
8 check for quit: if so exit
9 delete file
10 go to 5

R:
1 recover old command
2 jump to do-command

RENAME:
1 parse new filename
2 store this
3 parse old filename
4 check if old filename exists
5 error if it doesn't
6 replace old name with new name
7 exit

ROUTE:
1 parse for baud rate, word length, stop bits, parity
2 setup card if present
3 exit

SAVE:
1 parse filename
2 parse start, end address
3 check for execution address
4 if none then exec. address = end address
5 save file
6 exit

STMT:
1 get message from input
2 display message
3 exit

SYSTEM:
1 parse input
2 if null then display all the SYSTEM flags: exit
3 else check for get: if so display needed flag: exit
4 if set then parse input: put value to the flag: exit

TIME:
1 parse input
2 if null display current time: exit
3 else put new time in memory
4 exit

ACCESSING VDP REGISTERS (SV318/328)

To write to VDP register 7, first send the data to port &H81 then send the register number to this port —

```
10 OUT &H81,&B01001111
20 OUT & H81, (128 OR 7)
```

This performs the same function as COLOR 4,15. The left four bits of the binary number in line 10 are the foreground color and the other bits are background.

To read the VDP Status register use —

PRINT INP(&H85)

SV ROM DECODED (Part Two)

Spectravideo SV-318/328 ROM version 1.1

```
1374 LINE                  1CF0 RENUM
13A0 'Redo  from  start'   1E15 CSAVE
13B3 Dislay above msg      1E6E write HL  to  cass.
13CB INPUT#                1E76 read  HL from cass.
13D2 INPUT                 1E7F BLOAD (cassette)
1405 READ                  1EAA CLOAD
1495 'Extra ignored' msg   1F9F print filename
14CA Get prograam param.   1FA7 Cassette write
162D check for +-"& etc    2016 read  A from cass.
167B Functions             2026 write A to   cass.
16A2 VARPTR                203A Setup cass. for READ
16EF '-' Minus handling    204A Device I/O Error
170B Make upper case       2059 Setup cass. for WRITE
171A Base converter &H,    206C Write stop signal
     &O.&B to decimal      207C Stop motor
1834 LPOS                  2084 Check cassette
1839 POS                   209A 'Press play' message
183C Put A in L            20AF 'play and record' msg
183E Put A in H            20CF Cassette routines
1842 USR                   21C0 Handler for Macro
187B DEFUSR                     Languages (GML & MML)
188A DEF                   21F3 routines for above
18AD FN                    22CC exec macro substring
1A0E DEF FN                22DF get (X,Y) co-ord.
1A37 INP(x)                2328 PRESET (X,Y),C
1A4C OUT(x),y              232D PSET   (X,Y),C
1A52 WAIT                  2346 POINT  (X,Y)
1A6C WIDTH                 2390 Get color
1AA6 Get A entry 1         23B1 routines for above
1AA9 Get A entry 2         23E7 LINE(X,Y)-(X,Y),C,BF
1AB3 LLIST                 243C routines for above
1AB8 LIST                  2452   "     "    "
1C6C DELETE                247C   "     "    "
1CA6 PEEK                  249C   "     "    "
1CAD POKE                  24B0   "     "    "
1CB9 get DE,HL             24F4   "     "    "
```

***** The end of instalment 2 *****

MSX ROM CALLS

All MSX ROM calls should be made through the BIOS entry table.

MSX1 BIOS ENTRY TABLE

```
0000H  Name:      CHKRAM
       Function:  Check RAM/sets slots for command area
       Entry:     No parameters required before calling
       Returns:   No parameters are returned
       Modifies:  All registers

0008H  Name:      SYNCHR
       Function:  Checks if char pointed to by HL is
                  the one required.  If not, display
                  syntax error and jump to 0010H
       Entry:     HL, char to check is placed at 000BH
       Returns:   HL points to next char. Char in A
       Modifies:  AF, HL

000CH  Name:      RDSLT
       Function:  Selects slot according to value given
                  through registers and reads memory
                  contents from the slot.
       Entry:     A:  FxxxSSPP
                      F=1 if secondary slot specified
                      SS secondary slot 0-3
                      PP primary slot 0-3
                  HL: Target memory address
       Returns:   A:  Contents of memory location
       Modifies:  AF, BC, DE
       Note:      Interrupts are disabled

0010H  Name:      CHRGTR
       Function:  Gets next character (or token) from
                  BASIC
       Entry:     HL
       Returns:   HL=next chr addr., A has the char.
                  Sets carry flag if number, Z flag
                  set if end of statement.
       Modifies:  AF, HL

0014H  Name:      WRSLT
       Function:  Write version of RDSLT
       Entry:     Same as RDSLT with data to write in E
       Returns:   None
       Modifies:  AF, BC, D
       Note:      Interrupts are disabled.

0018H  Name:      OUTDO
       Function:  Output to current device
       Entry:     A, PTRFIL, PRTFLG
       Returns:   None
       Modifies:  None

001CH  Name:      CALSLT
       Function:  Inter-slot call to specified address
       Entry:     IY:  FxxxSSPP
                       F=1 if secondary slot specified
                       SS secondary slot 0-3
                       PP primary slot 0-3
                  IX:  Address to call
       Returns:   None
       Modifies:  None
       Note:      Interrupts disabled

0020H  Name:      DCOMPR
       Function:  Compares HL with DE
       Entry:     HL, DE
       Returns:   Flags
       Modifies:  AF

0024H  Name:      ENASLT
       Function:  Same as RDSLT except slot is perman-
                  ently enabled
       Entry:     Same as RDSLT
       Returns:   None
       Modifies:  All

0028H  Name:      GETYPR
       Function:  Returns type of floating point accum.
       Entry:     Floating point accumulator
       Returns:   Flags
       Modifies:  AF
```

COMPUTER KIDS

Hi kids! It's Wendy again in issue 5. Gee, doesn't time fly? It seems only days ago when I was writing the first Computer Kids column.

This issue, we've got another great word puzzle and a brilliant program that draws lots and lots of circles in all sorts of different colours.

If you're stuck for something to do in those long, boring hours in between meals, why not try playing computer games? You'll find lots of great game titles for sale for both the SV and MSX computers — and just in case your local computer shop doesn't have any, there's bound to be an advertiser in Computer Forum who does.

So, there's something you can save your pocket money for. Or, you can have it the easy way, ask for it for Christmas! Happy game playing!

Bye for now, Wendy.

```
10 REM CIRCLES OF COLOUR
20 COLOR 4,15
30 SCREEN 1 : REM FOR MSX
   USE SCREEN 2
40 C=1
50 FOR R=10 TO 100 STEP 8
60 FOR D=0 TO 7
70 CIRCLE(128,96),R+D,C
80 NEXT D
90 C=C+1
100 CLS
110 NEXT R
120 GOTO 10
```

Your word is: **COMPUTER FORUM**

PROGRAM LISTINGS

Each issue of COMPUTER FORUM will contain program listings for you to type in. When entering these programs it is a good idea to save off to cassette or disk every 10 or 20 lines and then continue. Then if the power goes off or another major disaster occurs, you will not have to type in the whole lot again.

Another check is to verify each save using the CSAVE? command. To use this command, CSAVE your program, then rewind the tape and type CLOAD? and press enter. This command will then compare the program on the tape with the one in memory, to verify that it was saved correctly.

All of the listings are printed in 40 column format. If you enter WIDTH40 before typing them in, you will be able to line up the characters in the listing with the text on screen.

Take care that you do not mistake the letter l for 1 or an O for a zero. Where graphics characters are used in the program, they will be shown as @1@. This symbol will then be repeated at the end of the listing with instructions concerning the graphics character used.

All of the listings have been tested on MSX and SV318/328 computers and do work. Should you have any problems, we suggest that you thoroughly check your work as errors can creep in very easily. Should you not wish to type in these listings, send a blank tape together with $2 to COMPUTER FORUM and we will post a copy to you. Do not forget to tell us which computer you have.

This program is suitable for use on SV318, SV328 and MSX.

EXTRA MORTGAGE PAYMENTS

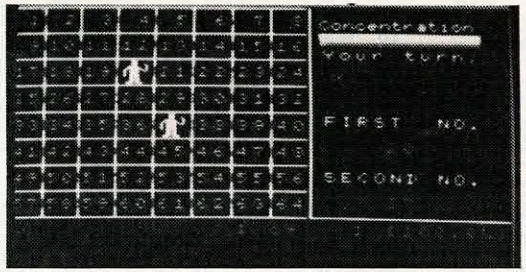
```
THIS PROGRAM CAN BE USED AS A GUIDE TO
CALCULATE THE EFFECT OF PAYING EXTRA
OFF THE MORTGAGE EACH MONTH.
THIS CAN ONLY BE APPROXIMATE AS BANKS AND
BUILDING SOCIETIES CALCULATION METHODS
WILL BE DIFFERENT.
```

```
100 CLS
110 PRINT SPACE$(8):"EXTRA MORTGAGE PAYM
ENTS"
120 PRINT:PRINT
130 INPUT "ENTER AMOUNT BORROWED":AB
140 OB=AB:NB=AB
150 PRINT
160 INPUT "ENTER INTEREST RATE EG 13.5":
IR
170 PRINT
```

```
180 INPUT "ENTER CURRENT PAYMENT";CP
190 PRINT
200 INPUT "ENTER PROPOSED PAYMENT";PP
210 CP$=STR$(CP):PP$=STR$(PP)
220 H1$="YEAR"+SPACE$(8)+"BALANCE"+SPACE
$(12)+"BALANCE"
230 H2$="No."+SPACE$(10)+"OWING"+SPACE$(
14)+"OWING"
240 H3$=SPACE$(11)+"@"+CP$+" /MTH"+SPACE
$(9)+"@"+PP$+" /MTH"
250 F$="###       ###,###.##          ###
,###.##"
260 GOSUB 440
270 I=IR/1200:.01
280 YR=YR+1
290 FOR M=1 TO 12:IP=IP+1
300 QI=NB*I*3
310 Q1=OB*I*3
320 IF IP=3 THEN NB=NB+QI-PP*3:QI=0:IP=0
:OB=OB+Q1-CP*3:T1=T1+CP*3
330 NEXT M
340 IF NB<0 THEN NB=0:GOTO 350 ELSE T2=T
2+PP*12
350 IF OB<0 THEN 410
360 PRINT USINGF$;YR,OB,NB
370 C=C+1:IF C<10 THEN 280
380 PRINT:PRINT "PRESS A KEY TO CONTINUE
"
390 I$=INKEY$:IF I$="" THEN 390
400 C=0:GOSUB 440:GOTO 280
410 PRINT "TOTAL"
420 PRINT "PAYMENTS";:PRINT USINGRIGHT$(
F$,31);T1,T2
430 END
440 CLS:PRINT "AMOUNT BORROWED =";AB
450 PRINT "INTEREST RATE =";IR
460 PRINT
470 PRINT H1$:PRINT H2$:PRINT H3$
480 PRINT STRING$(39,"_")
490 RETURN
```

☼ ☼ ☼

CONCENTRATION



NOTE: This listing is for MSX only. Due to it's complex nature, alterations are not given for Spectravideo.
If you want to try altering it you will need to deduct 1 from all SCREEN values and change all PRESETs to LOCATEs.

```
10 '-----------------------------------
20 '          CONCENTRATION
30 '    A game to test your memory
40 '
50 '    Author :   John Knight
60 '    Date   :   FEB  1986
70 '    Machine:   M.S.X (Spectravideo)
80 '    Memory :   16k   (approx req'd)
90 '-----------------------------------
100 DEFINTA-M,P-Z:KEYOFF:CLEAR1000
110 DIM PL(32),MN(64,3),SC(4,2)
120 '-----------------------------------
130 GOSUB710:'Title screen
140 GOSUB820:'Players names & level
150 GOSUB510:'Draw screen, etc
160 GOSUB650:'Setup sprites from data
170 GOSUB1000:'Put sprites into pool acc
ording to level number
180 GOSUB1070:'Move posns & sprite to ma
in array
190 '-----------------------------------
200 'MAINLINE ROUTINE
210 '-----------------------------------
220 FORNX=1TON
230 IF SC(1,1)+SC(2,1)+SC(3,1)+SC(4,1)>3
1THEN 450
```

```
240 LINE(160,44)-(252,60),1,BF:BEEP:PRES
ET(165,48),1:COLOR,11:PRINT#1,NM$(NX) .
250 GOSUB1170: 'guess 1 (g1)
260 X1=MN(G1,0):Y1=MN(G1,1):S1=MN(G1,2):
C1=MN(G1,3)
270 LINE(X1,Y1)-(X1+16,Y1+16),1,BF
280 S3=S1:GOSUB 1470:'sprite1 colour
290 PUT SPRITE0,(X1+1,Y1),C3,S1:GOSUB209
0:' sound
300 GOSUB1320: 'guess 2 (g2)
310 X2=MN(G2,0):Y2=MN(G2,1):S2=MN(G2,2):
C2=MN(G2,3)
320 LINE(X2,Y2)-(X2+16,Y2+16),1,BF
330 S3=S2:GOSUB 1470:'sprite2 colour
340 PUT SPRITE1,(X2+1,Y2),C3,S2:SC(NX,2)
=SC(NX,2)+1:GOSUB 2090:' sound
350 IF S1<>S2 THEN GOSUB 2000:GOTO 410
360 GOSUB 2170:' sound
370 SC(NX,1)=SC(NX,1)+1:GOSUB1660
380 MN(G1,2)=0:MN(G2,2)=0
390 GOSUB 1740:'clear square
400 GOTO 230:  'another go
410 GOSUB 1660:'print score
420 GOSUB 1740:'clear square
430 GOSUB 1850:'renumber square
440 NEXT NX:GOTO220
450 LINE(0,0)-(255,150),1,BF:   PRESET(7
0,100),1:PRINT#1,"** GAME OVER **"
460 GOSUB 2230
470 A$=INKEY$:IF A$="" THEN GOSUB2190:GO
TO 470 ELSE RUN
480 '-----------------------------------
490 'S U B R O U T I N E-S E C T I O N
500 '-----------------------------------
510 'DRAW SCREEN (for playing)
520 '-----------------------------------
530 COLOR7,1,4:SCREEN2,2
540 FORNN=1TO153STEP19:LINE(NN+2,1)-(NN+
2,153),7:LINE(1,NN)-(153,NN),7:NEXT:LINE
(155,153)-(255,153),7
550 A$="Concentration":COLOR5:FORI=1TOLE
N(A$):PRESET(154+(I*6),10):PRINT#1,MID$(
A$,I,1):NEXTI
560 LINE(160,20)-(240,22),11,BF
570 LINE(160,23)-(240,25),5,BF
580 LINE(160,26)-(240,28),6,BF
590 PRESET(162,32):PRINT#1,"Your turn,":
PRESET(162,80),11:PRINT#1,"FIRST  NO.":P
RESET(162,120),11:PRINT#1,"SECOND NO."
600 FORNN=1TON:PRESET(8,158+((NN-1)*8)),
1:COLOR4:PRINT#1,NM$(NN):NEXT:NN=1
610 FORI=0TO7:FORJ=0TO7:PRESET(6+(J*19),
7+(I*19)),1:COLOR6:PRINT#1,USING"##";NN:
NN=NN+1:NEXT:NEXT
620 PRESET(165,48),1:COLOR11:PRINT#1,"PL
EASE WAIT":COLOR6
630 RETURN
640 '-----------------------------------
650 'SETUP SPRITES
660 '-----------------------------------
670 LQ=L*8:IFLQ=24THENLQ=32
680 FORNN=1TOLQ:A$="":FORI=1TO32:READQ:A
$=A$+CHR$(Q):NEXT:SPRITE$(NN)=A$:NEXT
690 RETURN
700 '-----------------------------------
710 'TITLE SCREEN
720 '-----------------------------------
730 SCREEN3,2:OPEN"grp:"AS1:COLOR4,1,4:C
LS
740 A$="CONCENTRATION":NN=RND(-TIME)
750 FORI=1TOLEN(A$):NN=FIX(RND(1)*14)+2:
PRESET(36+(I*12),50),1:COLORNN:PRINT#1,M
ID$(A$,I,1):B$=STR$(60+I):B$="V3 m03000s
12t255n"+B$:PLAYB$:FORN8=1TO90:NEXTN8:CL
S:NEXT
760 FORNN=1TO800:NEXT
770 SCREEN2,3
780 FORNN=1TO32:A$="":FORI=1TO32:READQ:A
$=A$+CHR$(Q):NEXT:SPRITE$(1)=A$:S3=NN:GO
SUB1490
790 PUT SPRITE0,(FIX(RND(1)*200)+10,FIX(
RND(1)*150)+10),C3,1:GOSUB2090:FORJ=1TO1
00:NEXT:NEXT:RESTORE
800 RETURN
810 '-----------------------------------
820 'GET PLAYERS AND LEVEL
830 '-----------------------------------
840 COLOR7,1:CLS:SCREEN1:PRINT"  Test yo
ur memory by wiping out the board in as
few movesas possible. Note that screense
tup  may  take  up  to  15 seconds."
850 PRINT:PRINT:INPUT"How many players (
1-4)";N
860 IFN<1ORN>4THEN840
870 FORI=1TON:PRINT:PRINT"Name of player
 ";I
880 INPUTNM$(I):IFLEN(NM$(I))>10THENNM$(
I)=LEFT$(NM$(I),10)
```

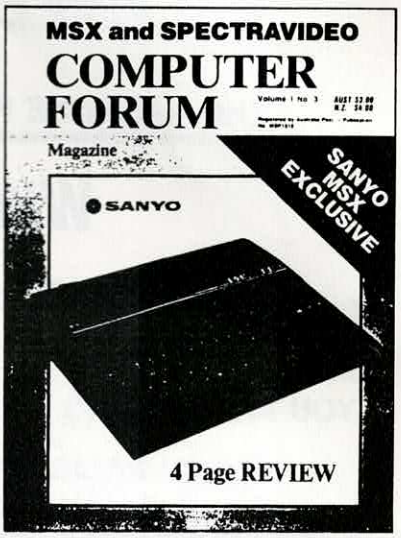
```
890 NEXT:PRINT:PRINT"CHOOSE DIFFICULTY L
EVEL (1-3)":PRINT"     1. EASY":PRINT"
   2. INTERMEDIATE":PRINT"     3. DIFFIC
ULT"
900 B$=INKEY$:IFB$=""THEN900
910 IFB$<"1" OR B$>"3" THEN 900
920 L=VAL(B$):IF L=3 THEN 980
930 IF L=2 THEN NN=FIX(RND(-TIME))*2:IF
NN=2 THENRESTORE2510:GOTO980ELSEGOTO980
940 IF NN=1 THEN 980
950 IF NN=2 THEN RESTORE2430:GOTO980
960 IF NN=3 THEN RESTORE2510:GOTO980
970 RESTORE2590
980 RETURN
990 '-----------------------------------
1000 'PUT SPRITES INTO POOL ACCORDING TO
 DIFFICULTY LEVEL
1010 '-----------------------------------
1020 IFL=1THENFORI=1TO8:PL(I)=8:NEXT:GOT
O1050
1030 IFL=2THENFORI=1TO16:PL(I)=4:NEXT:GO
TO1050
1040 FORI=1TO32:PL(I)=2:NEXT
1050 RETURN
1060 '-----------------------------------
1070 'RANDOMISE TO ARRAY AND INCLUDE THE
 SPRITE POSITIONS
1080 '-----------------------------------
1090 ZZ=RND(-TIME):IF L=3 THEN L=4
1100 N2=0:FORI=0TO7:FORJ=0TO7:N2=N2+1
1110 NN=FIX(RND(1)*8*L)+1: IFPL(NN)=0THE
N1110
1120 MN(N2,0)=J*19+4:MN(N2,1)=I*19+2:MN(
N2,2)=NN:PL(NN)=PL(NN)-1
1130 MN(N2,3)=11
1140 NEXT:NEXT
1150 RETURN
1160 '-----------------------------------
1170 '       GET_A_GUESS (1)
1180 '-----------------------------------
1190 PRESET(192,98),1:D1=0:D2=0:FLAG=0
1200 G$=INKEY$:IFG$=""THEN1200
1210 IFG$=CHR$(13)THEN1260
1220 IF G$<"0" OR G$>"9" THEN BEEP:BEEP:
BEEP:GOTO1250
1230 IFFL=0ANDVAL(G$)=>0ANDVAL(G$)<=9THE
ND1=VAL(G$):FL=1:PRINT#1,G$;:GOTO1200
1240 IFFL=1ANDVAL(G$)=>0ANDVAL(G$)<=9THE
ND2=VAL(G$):FL=2:PRINT#1,G$;:GOTO1200
1250 LINE(190,96)-(220,110),1,BF:GOTO119
0
1260 IF FL=1 THEN G1=D1:GOTO1280
1270 G1=(10*D1)+D2
1280 IF G1<1 OR G1>64 THEN 1250
1290 IF MN(G1,2)=0 THEN 1250
1300 RETURN
1310 '-----------------------------------
1320 '       GET_A_GUESS (2)
1330 '-----------------------------------
1340 PRESET(192,138),1:D1=0:D2=0:FLAG=0
1350 G$=INKEY$:IFG$=""THEN1350
1360 IFG$=CHR$(13)THEN1410
1370 IF G$<"0" OR G$>"9" THEN BEEP:BEEP:
BEEP:GOTO1400
1380 IFFL=0ANDVAL(G$)=>0ANDVAL(G$)<=9THE
ND1=VAL(G$):FL=1:PRINT#1,G$;:GOTO1350
1390 IFFL=1ANDVAL(G$)=>0ANDVAL(G$)<=9THE
ND2=VAL(G$):FL=2:PRINT#1,G$;:GOTO1350
1400 LINE(190,136)-(220,150),1,BF:GOTO13
40
1410 IF FL=1 THEN G2=D1:GOTO1430
1420 G2=(10*D1)+D2
1430 IF G2<1 OR G2>64 THEN 1400
1440 IF MN(G2,2)=0 OR G1=G2 THEN 1400
1450 RETURN
1460 '-----------------------------------
1470 '  SELECT COLOR FOR SPRITE
1480 '-----------------------------------
1490 IF S3=1ORS3=15ORS3=22ORS3=26THEN C3
=11:GOTO1640
1500 IF S3=2ORS3=9ORS3=17 THEN C3=13:GOT
O1640
1510 IF S3=3 THEN C3=9:GOTO1640
1520 IF S3=4ORS3=11ORS3=18ORS3=31THEN C3
=15:GOTO1640
1530 IF S3=5ORS3=12ORS3=21ORS3=24ORS3=30
THEN C3=7:GOTO1640
1540 IF S3=27ORS3=28THEN C3=8:GOTO1640
1550 IF S3=23ORS3=32THEN C3=5:GOTO1640
1560 IF S3=6 THEN C3=6:GOTO1640
1570 IF S3=20THEN C3=3:GOTO1640
1580 IF S3=7 THEN C3=4:GOTO1640
1590 IF S3=8ORS3=29THEN C3=10:GOTO1640
1600 IF S3=16ORS3=10ORS3=14THEN C3=14:GO
TO1640
1610 IF S3=19ORS3=25THEN C3=12:GOTO1640
1620 IF S3=13 THEN C3=2:GOTO1640
1630 C3=15
1640 RETURN
1650 '-----------------------------------
1660 '     PRINT SCORE
```

```
1670 '-----------------------------------
1680 LINE(104,150+(NX*8))-(254,157+(NX*8
)),1,BF
1690 PRESET(112,150+(NX*8)),1
1700 PRINT#1,USING"##";SC(NX,1);:PRINT#1
," of ";
1710 PRINT#1,USING"###";SC(NX,2);:PRINT#
1," [";
1720 PRINT#1,USING"###.#";100*SC(NX,1)/S
C(NX,2);:PRINT#1,"%]":RETURN
1730 '-----------------------------------
1740 '     BLANK OUT BOTH SQUARES
1750 '-----------------------------------
1760 FORNN=1TO900:NEXTNN
1770 LINE(X1+1,Y1)-(X1+16,Y1+16),1,BF
1780 LINE(X2+1,Y2)-(X2+16,Y2+16),1,BF
1790 PUT SPRITE0,(0,209)
1800 PUT SPRITE1,(0,209)
1810 LINE(190,96)-(220,110),1,BF
1820 LINE(190,136)-(220,150),1,BF
1830 RETURN
1840 '-----------------------------------
1850 '     RE-ESTABLISH BOX NUMBER
1860 '-----------------------------------
1870 COLOR,4:PSET(X1+2,Y1+5),1:PRINT#1,U
SING"##";G1
1880 PSET(X2+2,Y2+5),1:PRINT#1,USING"##"
;G2
1890 RETURN
1900 '-----------------------------------
1910 ' SOUND AT GAME_OVER
1920 '-----------------------------------
1930 SOUND7,190:SOUND8,8
1940 FORI%=1TO15:FORNJ=1TO10:SOUND1,I%:N
EXT:TIME=0
1950 IF TIME<1THEN1950
1960 NEXT:SOUND8,0:RETURN
1970 '-----------------------------------
1980 'SOUND FOR "NO_MATCH"
1990 '-----------------------------------
2000 TIME=0
2010 IF TIME<20  THEN 2010
2020 SOUND7,182:SOUND8,9 :SOUND1,15:TIME
=0
2030 IF TIME<20 THEN 2030
2040 SOUND8,0
2050 RETURN
2060 '-----------------------------------
2070 'SOUND on SPRITE DISPLAY
2080 '-----------------------------------
2090 SOUND7,190:SOUND8,12
2100 FORI%=15TO1STEP-2:SOUND1,I%:TIME=0
2110 IF TIME<1THEN2110
2120 NEXT:SOUND8,0
2130 RETURN
2140 '-----------------------------------
2150 'SOUND FOR "MATCH"
2160 '-----------------------------------
2170 PLAY"t155L64o6R42cegR10eggg"
2180 RETURN
2190 '-----------------------------------
2200 '    END_ROUTINE
2210 '-----------------------------------
2220 NN=FIX(RND(1)*15)+1:LINE(50,80)-(21
0,128),NN,BF:NN=FIX(RND(1)*15)+1:PRESET(
70,100),NN:COLORNN:PRINT#1,"** GAME OVER
 **":GOSUB1930:FORNN=1TO100:NEXT:RETURN
2230 PRESET(30,2),0:COLOR13:PRINT#1,"GAM
E RATINGS  - LEVEL ";L:LINE(30,15)-(230,
15)
2240 IFL=1 THEN S1=75:S2=60:S3=40:S4=30
2250 IFL=2 THEN S1=70:S2=55:S3=35:S4=25
2260 IFL=3 THEN S1=65:S2=50:S3=30:S4=20
2270 PRESET(30,20),0:PRINT#1,"UNBELIEVAB
LE  - Over ";USING"## %";S1
2280 PRESET(30,30),0:PRINT#1,"EXCELLENT
   - Over ";USING"## %";S2
2290 PRESET(30,40),0:PRINT#1,"GOOD
    - Over ";USING"## %";S3
2300 PRESET(30,50),0:PRINT#1,"FAIR
   - Over ";USING"## %";S4
2310 RETURN
2320 '-----------------------------------
2330 'SPRITE DATA -- The following secti
on contains the sprite data. There are 3
2 16x16 sprites. Each is named.
2340 '-----------------------------------
2350 DATA38,38,38,63,63,63,15,15,15,11,1
1,15,15,14,12,28,200,200,200,248,248,248
,224,224,224,160,160,224,224,224,96,112:
' tower
2360 DATA48,112,224,192,192,195,197,255,
255,197,195,192,192,224,112,48,24,28,14,
6,6,134,70,254,254,70,134,6,6,14,28,24:'
 tighe fighter
2370 DATA2,4,68,67,69,71,35,151,142,124,
30,124,142,151,35,32,128,64,68,132,68,19
6,136,210,226,124,240,124,226,210,136,8:
' spider
2380 DATA0,7,15,31,63,63,51,51,19,30,12,
15,8,8,7,3,0,192,224,240,248,248,152,152
,144,240,96,224,32,32,192,128:' skull
```

```
2390 DATA0,5,3,3,4,8,16,16,16,16,16,8,4,
3,0,0,0,160,192,192,32,16,8,8,8,8,8,16,3
2,192,0,0:' diamond ring
2400 DATA24,36,68,136,156,110,7,3,1,0,0,
0,1,3,1,0,0,0,0,0,0,0,0,128,192,224,112,
248,124,236,64,128:' key
2410 DATA0,1,3,6,12,24,49,99,199,99,49,2
4,12,6,3,1,0,0,128,192,96,48,24,140,198,
140,024,48,96,192,128,0:' double diamond
2420 DATA3,3,3,1,15,23,39,39,39,39,38,6,
6,6,6,30,128,128,136,4,196,228,212,200,1
92,192,192,192,192,192,192,240:' man
2430 DATA136,200,165,147,143,131,131,127
,39,67,67,79,91,115,97,193,34,38,74,146,
226,130,130,252,136,132,132,228,180,156,
12,6:' butterfly
2440 DATA0,1,27,31,31,31,63,127,192,120,
75,74,122,66,66,127,0,0,128,192,224,240,
248,252,6,60,164,164,188,132,132,252:' h
ouse
2450 DATA0,0,0,0,0,0,64,103,63,31,15,15,
7,1,1,1,0,26,28,18,16,16,16,136,200,248,
248,240,224,0,0,192:' duck
2460 DATA0,0,0,32,112,248,223,143,7,7,8,
16,16,8,0,0,0,0,0,0,2,5,249,249,249,241,
8,8,16,32,0,0:' horse
2470 DATA0,0,0,0,15,18,34,255,255,255,25
5,56,0,0,0,0,0,0,0,0,192,32,16,254,254,2
54,254,28,0,0,0,0:' car
2480 DATA0,120,120,48,49,51,255,255,255,
255,128,255,255,247,99,0,0,0,31,31,24,15
2,248,248,248,248,24,255,255,188,24,0:'
locomotive
2490 DATA0,0,63,64,127,64,64,64,70,73,73
,70,64,64,32,31,0,0,192,32,224,60,34,34,
34,36,40,48,32,32,64,128:' mug
2500 DATA0,3,3,3,3,31,31,255,255,255,255
,127,0,0,0,0,0,0,0,0,0,8,208,224,255,254,2
52,248,240,0,0,0,0:' battleship
2510 DATA0,0,0,0,255,127,63,31,255,255,1
15,113,80,0,0,0,0,0,56,112,224,192,192,1
92,192,192,192,192,192,64,0,0:' plane
2520 DATA1,1,41,19,41,7,21,255,20,7,41,1
9,41,1,1,0,0,0,40,144,40,192,80,254,80,1
92,40,144,40,0,0,0:' snow flake
2530 DATA15,16,39,72,147,164,169,171,171
,169,164,147,72,39,16,15,224,16,200,36,1
46,74,42,170,170,42,74,146,36,200,16,224
:' bullseye
2540 DATA12,8,15,16,32,70,64,65,80,79,72
,39,16,15,3,3,48,16,240,8,4,98,2,130,10,
242,18,228,8,240,192,192:' devil
2550 DATA0,0,0,1,1,1,1,3,7,14,28,56,112,
224,192,0,32,64,136,16,162,196,232,240,0
,0,0,0,0,0,0,0,0:' fork
2560 DATA3,7,3,3,130,66,36,8,16,208,16,1
6,8,36,67,128,192,224,192,192,65,66,36,1
6,8,11,8,8,16,36,194,1:' elec.light
2570 DATA255,8,28,8,28,34,67,131,135,255
,255,127,34,162,127,0,0,0,0,0,2,5,2,255,
255,254,248,0,0,64,128,0:' chopper
2580 DATA0,0,0,7,15,24,49,98,98,49,24,15
,7,0,0,0,0,0,0,224,240,24,140,70,70,140,
24,240,224,0,0,0:' eye
2590 DATA0,0,32,64,128,128,255,224,224,2
55,64,224,224,224,0,0,0,0,0,0,0,0,255,1,
1,255,0,0,0,0,0,0:' hammer
2600 DATA255,192,191,176,175,172,171,171
,171,171,172,175,176,191,192,255,255,3,2
53,13,245,53,213,213,213,213,53,245,13,2
53,3,255:' squares
2610 DATA16,24,20,242,74,38,30,1,0,0,0,0
,0,0,0,0,0,0,0,0,0,0,0,0,128,64,32,17,11
,7,7,15:' arrow
2620 DATA1,2,2,15,31,63,111,255,255,127,
7,31,15,2,2,1,0,0,0,128,193,227,247,255,
247,227,193,128,0,0,0,0:' fish
2630 DATA12,12,12,15,12,12,12,15,12,12,1
2,15,12,12,12,12,48,48,48,240,48,48,48,2
40,48,48,48,240,48,48,48,48:' ladder
2640 DATA1,2,0,0,0,0,1,2,5,10,21,42,84,1
68,208,224,131,199,110,60,88,172,86,163,
65,130,0,0,0,0,0,0:' sword
2650 DATA42,53,42,53,42,53,42,53,42,32,3
2,32,32,32,32,32,168,80,168,80,168,80,16
8,80,168,0,0,0,0,0,0,0:' cheq'd flag
2660 DATA0,0,0,112,32,16,28,24,20,50,89,
149,191,133,72,48,0,0,0,0,0,0,0,56,16,60
,82,169,253,161,18,12:' bicycle
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